

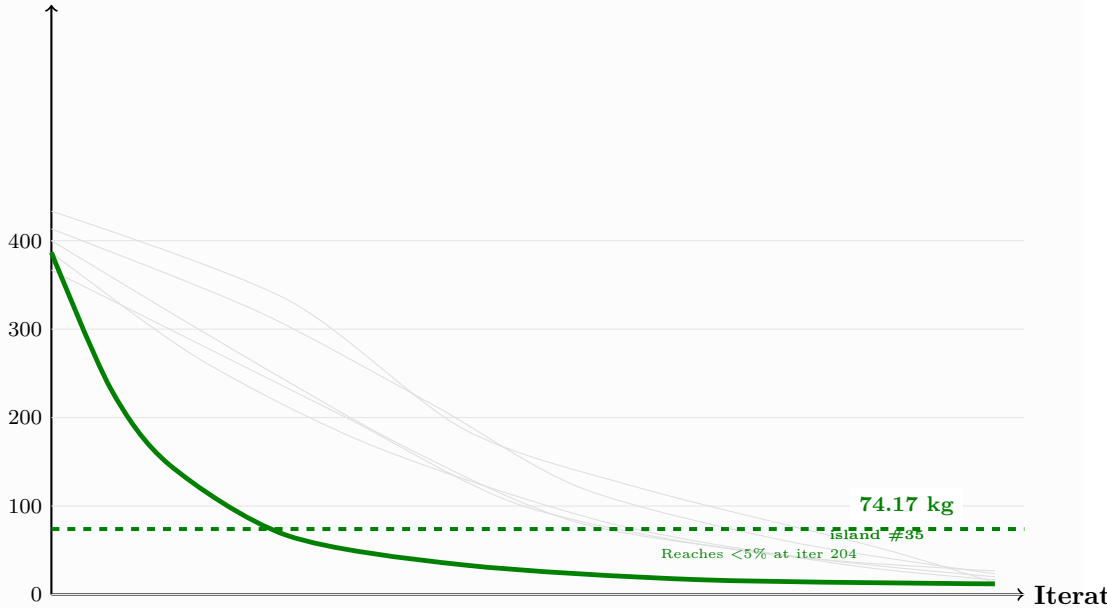
V6.2 Optimization Landscape

60 islands 600,000 evaluations 58/60 converged to 70-75 kg

1. Convergence: 58/60 Islands → 74.17 kg

Grey = individual islands. Green = best island (#35).

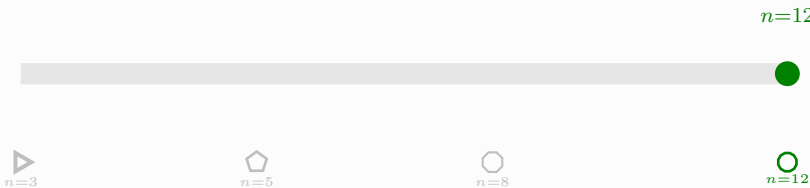
Best mass (kg)



Bounds: $n \in [3, 12]$, $r_{hub} \in [1, 10]m$, $r_{bot} \in [0.1, 5]m$,
 $\beta \in [-0.8, 0.8]$, $n_{rings} \in [5, 16]$, $n_{exp} \in [0, 6]$, $bank \in [0, 60]^\circ$

2. Polygon Search: Converged to $n=12$

Optimizer freely varied $n \in [3, 12]$ alongside 10 other parameters



Higher $n \rightarrow$ thinner beams $\rightarrow$ smaller knuckles

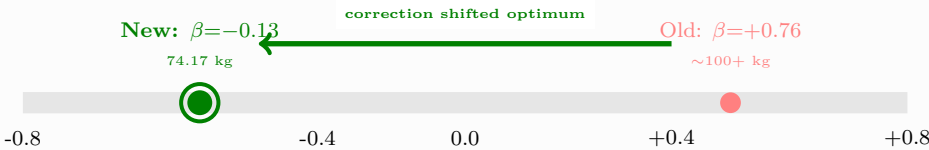
$n=12$ dodecagon: most efficient polygon in search range

Single-parameter sweep infeasible – optimum is a constraint intersection

58/60 islands independently converged to $n=12$

3. Density Profile: β Sign Flip

$\tan \rightarrow \sin + \cos^3 \rightarrow \cos^{2.65}$ corrections shifted optimum from $\beta=+0.76$ to $\beta=-0.13$



top-bias ($\beta < 0$)

uniform ($\beta = 0$)

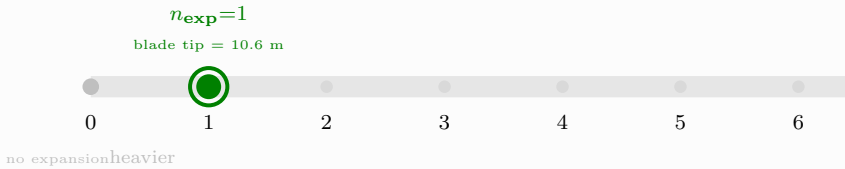
bottom-heavy ($\beta > 0$)

β controls ring spacing: $\beta > 0$ clusters toward bottom (high compression), $\beta < 0$ toward top (beam taper dominates)

Single-parameter β sweep infeasible – constraint intersection locks β to campaign value

4. Expansion Stations: $n_{exp}=1$ Optimal

Optimizer varied $n_{exp} \in [0, 6]$ AND blade radius $\in [0, 15]m$ independently



hub only

Each extra station adds: 3 blades + knuckle hardware + parasitic drag
The optimizer could use many small rotors or one large one – it chose one large rotor

Single-parameter sweep infeasible. Optimizer freely explored full 11-D space